

Optimal schedule recovery for the aircraft gate assignment with constrained resources

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ABSTRACT

Efficient ground handling at airports contributes significantly to the performance of the entire air transport network. In this network, airports are coupled by flights that depend on passenger and crew connections, effective local airport operations, and efficient ground handling resource management. In addition, airport stakeholders must consider different time scales (look-ahead times), process estimates, and both bounded and multiple-dependent solution spaces for their decision-making processes. In this context, we aim to solve the NP-hard problem of combined aircraft turnaround time and airport gate allocation optimization. We developed a hybrid meta-heuristic algorithm based on the Shuffled Frog-Leaping Algorithm (SFLA) and the Grasshopper Optimization Algorithm (GOA). Our hybrid SFLA-GOA approach overcomes the drawback of each algorithm and achieves a preferred solution at a reduced computational cost. We test our approach with a scenario derived from Frankfurt Airport operations, including gate positions, terminal layout, and flight plan information. In all defined problem sizes, the hybrid SFLA-GOA algorithm outperforms an independent implementation of the SFLA algorithm. Also, the SFLA-GOA algorithm provides an improved solution by at least nearly 12% up to 40% over the standard mixed-integer programming solvers.

1. Introduction

The aircraft turnaround is defined by the needed operational ground handling processes at the aircraft stand (position at the gate or the apron). The turnaround time is measured as the time between the in-block (aircraft stops and wheels are locked) and off-block time (wheels are unlocked). The turnaround consists of up to twelve interdependent sub-processes, containing more than 150 individual activities and involving up to 30 different actors (IATA, 2018). The most critical sub-processes are the processes that are part of the turnaround's critical path and directly influence the aircraft turnaround time.

The ground handling processes are related to catering, cleaning, fueling, baggage handling, and passenger handling. Whereas passenger handling requires no additional infrastructure or special support at gate position (except passengers with reduced mobility or children traveling alone), ground processes depend on infrastructure and resources, such as fuel trucks or ground power units. In the case of apron position, passengers have to be transported to the gate with buses. After the aircraft reaches the assigned stand, both local facilities and apron-wide available resources are used to start with the ground processes. Fig. 1 emphasizes the standard turnaround progress and a typical ramp layout

at an open apron position (Airbus, 2020).

Operational planning at the airport is challenged by deviations in aircraft operations (delay), local airport capacities (arrival and departure flow), local weather conditions, or resource availability. Typically variations from the flight plan may be separated into deviations (statistical, known variations), disturbances (unknown in time and severity, but part of standard procedures), and interruptions (changes to contingency procedures). Depending on the time horizon, operators from airlines, airports, and ground handling agencies can appropriately react. In this context, the airport collaborative decision making (A-CDM) enables the decision maker to share information between stakeholders of the complex airport system to provide an everyday situational awareness and to enable joined strategies to solve operational challenges (Eurocontrol Airport CDM Team, 2017). During the day of operations, updates are provided for each flight with time-based estimations for milestones along the aircraft trajectory, such as estimated times for arrival (see Fig. 2).

In Section 2 the related literature is reviewed to find the research gaps. Section 3 describes the problem and network of turnaround and mathematical formulation, while Section 4 presents a hybrid meta-heuristic algorithm SFLA-GOA, compares the performance of the

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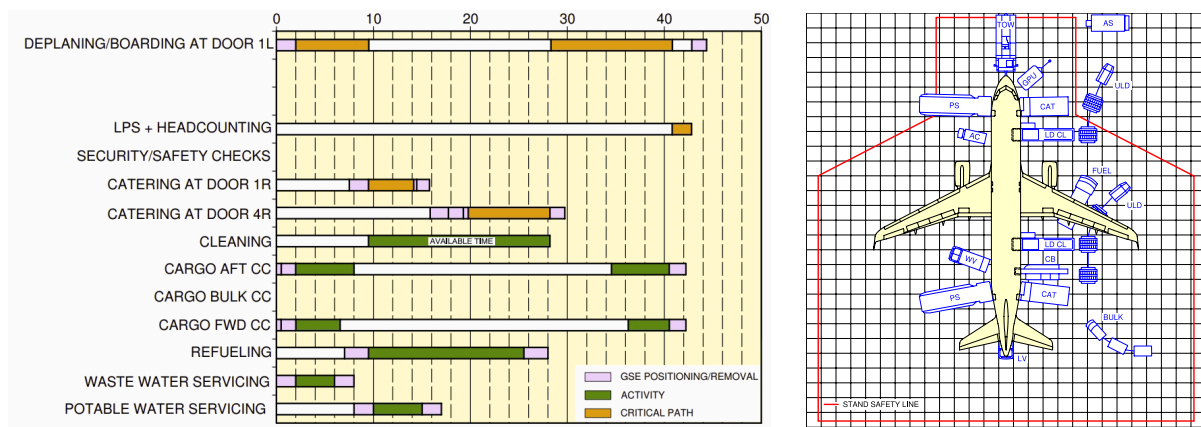


Fig. 1. (left) Standard turnaround time schedule considering time for ground support equipment (GSE) and critical path, (right) layout of Airbus A320 ground operations (Airbus, 2020).

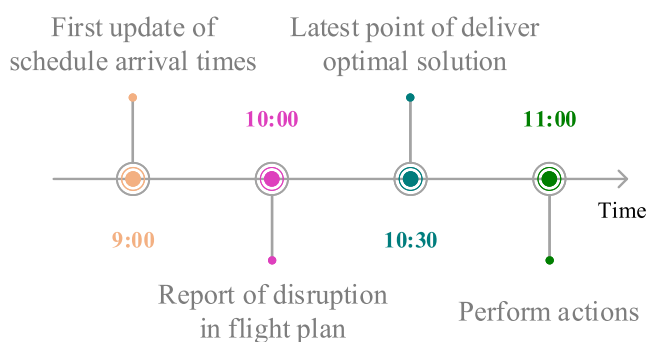


Fig. 2. Timeline of a single aircraft trajectory to enable decision making in the Airport Operations Centre (APOC, see [SESAR Joint Undertaking \(2016\)](#)).

proposed algorithm with a well-known (Shuffled Frog-Leaping Algorithm) SFLA algorithm. Section 5 presents a case study of twenty aircraft turnarounds during the morning peak at Frankfurt Airport using thirteen available gates. Furthermore, the performance of the hybrid algorithm is examined. Finally, in Section 6 the conclusion and an outlook on further researches is provided.

2. Literature review

One of the first models to find optimal recovery actions to minimize the passenger delay cost and integrate crew connection was proposed by [Bratu and Barnhart \(2006\)](#). They developed a bi-objective mathematical model to facilitate decision-making in an integrated area and inspect the interaction between turnaround activities in terms of time windows. A new heuristic algorithm called Sequence Iterative Method (SIM), with a combination of Constraint Programming (CP) and Large Neighborhood Search (LNS) and Variable Neighborhood Descent (VND), was presented to address their multi-objective problem. [Schmidt \(2017\)](#) introduced aircraft ground operations focusing on turnaround and passenger processes and analyzed the issues of the existing operations, namely airport capacity constraints, schedule disruptions, and the increasing cost pressure. [Kohl, Larsen, Larsen, Ross, and Tiourine \(2007\)](#) categorized different kinds of airline disruptions to ease reaction processes and reviewed existing methods which use a resource decision support system, to deal with disruptions in airline management. [Norin, Granberg, Yuan, and Värbrand \(2012\)](#) proposed a simulation model to optimize the schedule of turnaround process with a focus on De-Icing operation. The model implemented four different scenarios at Stockholm Arlanda airport with the real data to analyze the effect of the proposed model. Extending the model of their former research ([Padrón & Guimaraens,](#)

2020; Padrón, Guimarães, Ramos, & Fitouri-Trabelsi, 2016) developed a bi-objective mathematical model that describes the ground handling scheduling problem and considers time windows for each turnaround activity. In this approach, first, the model searches in the solution area to find the reliable answers using a heuristic algorithm and then narrows the solution space in the vicinity of the Pareto solution to find an optimum solution. This approach outperforms the previous method introduced by the authors. Jacquillat and Odoni (2015) developed a mathematical model which optimizes the decisions taken at both strategic and tactical levels. The proposed model is comprised of three main parts: (1) model airport congestion by a stochastic queuing model, (2) capacity utilization is optimized by a dynamic programming model, and (3) integer programming model for scheduling interventions. Thengvall, Bard, and Yu (2000) developed a scheduling model to find the best recovery actions when the scheduled operation plan deviates from its standard routing procedure because of disruptions. The model depicts the situation in the whole routing network. The model was solved by LP relaxation method. Wu, Wu, Feng, Zhang, and Qiu (2018) analyzed arrival delays and their effect on departure delays throughout the air traffic network (delay propagation). The distribution function of departure delay and the knock-on effects (correlation of delay propagation) is estimated and fitted by a copula function. To make flight schedules more reliable and avoid severe change of the schedule in delay conditions, different scenarios for buffer consideration are implemented. Al-Bazi, Gok, Ozturk, and Guimarães (2016) developed a mathematical model that optimized aircraft turnaround time focusing on low-cost carriers. The objective function of the proposed scheduling model is the completion time of the aircraft turnaround which should be minimized using the limited resources that exist in the airports. To optimize resource utilization for turning aircraft around and disruption management, Kuster, Jannach, and Friedrich (2009) introduced a new RCPSP model, called x-RCPSP. x-RCPSP model finds the optimal resource allocation plan for all serialized or parallel activities. It also considers multiple modes for each turnaround operation, which helps decide whether the operation should be shifted to another starting time or be served with more resources to compensate for the disturbance effect. A genetic algorithm was implemented to solve this NP-Hard problem efficiently with a reasonable solving time. Schultz (2013) introduced a framework that helps turnaround management for high automation levels by turnaround time reduction in case of disturbances. The results showed that aircraft cleaning and boarding processes have enormous potential for minimizing aircraft time. Vidosavljevic and Tosić (2010) developed an analytical model to minimize the total cost of the operating system in the area of aircraft turnaround among multiple turnarounds. This paper developed a mathematical model considering the stochastic nature of arrival and departure delays and analyzed the role of buffer in system reliability. Wu et al. (2004) developed an aircraft



Fig. 3. Picture of the relation between different resource handling in the airport environment on aircraft (microscopic) and airport (macroscopic) level.

rotation model in the network by aggregating two well-known sub-models, turnaround model and enroute model. The turnaround model focuses on the activities between in-block time and off-block time, and the enroute model simulates flight times between two airports. [Montlaur and Delgado \(2017\)](#) studied reactionary delays for flights and passengers integrating the ground holding problem.

Gate assignment is the process of selecting appropriate aircraft positions at the airport gates or on the apron to ensure short transfer times for passengers ([Ali, Guleria, Alam, & Schultz, 2019](#)) and optimal use of ground handling resources. Also, changes in operational procedures (e. g., delays) should only have a minor impact on the actual assignment of aircraft positions (robustness). [Cheng \(1998\)](#) proposed one of the first studies to model gate assignment in an airport. In this study, integration of a knowledge-based system and mathematical programming was proposed to find an optimal solution of assigning a group of aircraft to available gates. The model considers different objectives such as minimizing delay and waiting time, maximizing the use of fixed gates, minimizing the deviation of the pre-schedule assignment, minimizing passenger walk distance. [Xu and Bailey \(2001\)](#) proposed a mixed quadratic integer problem to minimize passenger walk time between their flights. Tabu Search algorithm was implemented to solve this NP-hard class problem. To enhance the tabu search method, they proposed some pre-calculation to provide improved lists. [Drexel and Nikulin \(2008\)](#) developed a multi-objective model that simultaneously minimizes passenger walk-time and maximizes gate assignment preferences. A Pareto simulated annealing was developed in order to solve this hard computational problem. [Genç, Erol, Eksin, Berber, and Güleriyüz \(2012\)](#) developed a model to maximize gate employment with a stochastic approach. To find the solution to this problem, a heuristic method called Big Bang-Big Crunch optimization was implemented to produce a good initial solution for a stochastic algorithm. [Guépet, Acuna-Agost, Briant, and Gayon \(2015\)](#) developed a mixed-integer model that integrates

different topics of aircraft activity (such as arrival and departure) with parking stands (gates), intending to maximize the number of aircraft on stands rather than remote positions. Since the proposed model belongs to the NP-hard classes, a heuristic algorithm was used to find the optimal solution to the problem. [Deng et al. \(2017\)](#) minimized the walking distance of passengers by developing a mathematical model. An improved version of PSO (called DOADAPO) algorithm was developed to solve the problem. The same problem was developed by [Aktel, Yagmahan, Özcan, Yenisey, and Sansarci \(2017\)](#) solved by Tabu Search (TS) algorithm and Simulated Annealing (SA). There are other studies that pursued identical approach by using different algorithms (see [Bolat, 2001; Ding, Lim, Rodrigues, & Zhu, 2005; Hu & Di Paolo, 2009; Xu & Bailey, 2001; Lim, Rodrigues, & Zhu, 2005](#)).

This paper integrates two main challenges of airlines and airports into an integrated optimization model: aircraft turnaround time and gate assignment. Because of the resemblance between turnaround time optimization and RCPSP problems, the turnaround time optimization can be formulated using RCPSP formulation ([Asadi, Evler, Preis, & Fricke, 2020](#)). Since RCPSP problem belongs to the class of NP-hard problems ([Gonzalez-Pardo, Del Ser, & Camacho, 2017; Habibi, Barzin-pour, & Sadjadi, 2017; Pellerin, Perrier, & Berthaut, 2020; Roy & Sen, 2019](#)) and as it can be concluded from the researches as mentioned earlier that gate assignment problems belong to NP-hard class, the combination of these two belongs consequently to the NP-Hard class of problems.

Concluding from the literature review and to the best of our knowledge, there has been no research concept so far which has been developed a solution method to solve the NP-Hard problem of combining turnaround and gate assignment. Understanding the full potential of schedule recovery and gate assignment during ground operations is an essential prerequisite for developing holistic decision support systems. The solution method presented in this article aims at

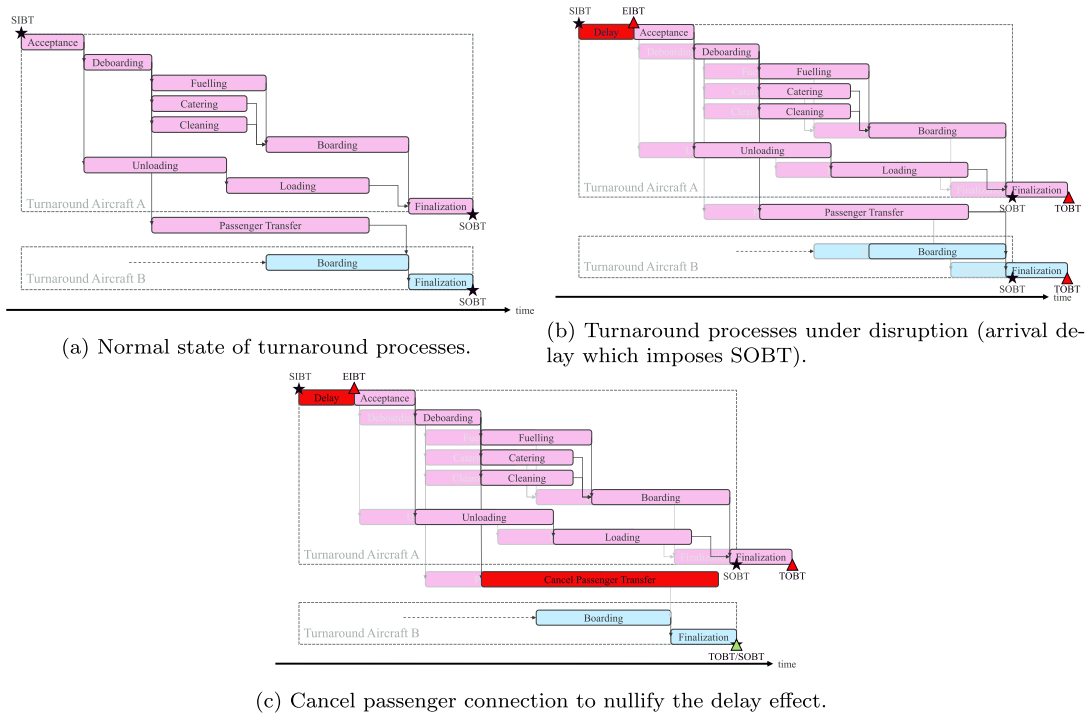


Fig. 4. Turnaround processes considering (a) normal and (b) disrupted conditions as well as (c) canceled passenger connections.

filling this gap in the literature by combining two different well-known meta heuristic algorithms SFLA and GOA and developing a new novel hybrid algorithm that has not been introduced before.

3. Problem statement

During the aircraft turnaround, the ground handler (contracted by the airline) performs the necessary aircraft processes handling passengers, baggage, cargo, and aircraft services. Thus, the ground handler organizes the required equipment, such as passenger stairs, push-back tugs, or fuel trucks (see Fig. 3). Resource optimization depends on the number of parallel handled aircraft and their dependencies (e.g., transfer passengers), the distance between (active) stands, the number of available resources, and in particular, the operational changes due to

delayed operations. Thus, ground handling has to consider microscopic (on aircraft level) and macroscopic (on airport level) constraints on the day of operations.

3.1. Turnaround network model

As stated in the International Air Transport Association (IATA) ground operations manual, the turnaround consists of twelve main processes that can split into up to 150 different sub-tasks (Asadi et al., 2020). Parallel turnaround operations are one of the main challenges in both airline and airport views. According to data from the summer schedule 2017 of the local hub carrier Lufthansa there could be up to 50 turnarounds simultaneously at an airport during peak time. Therefore, predicting the accurate completion time turnaround processes and

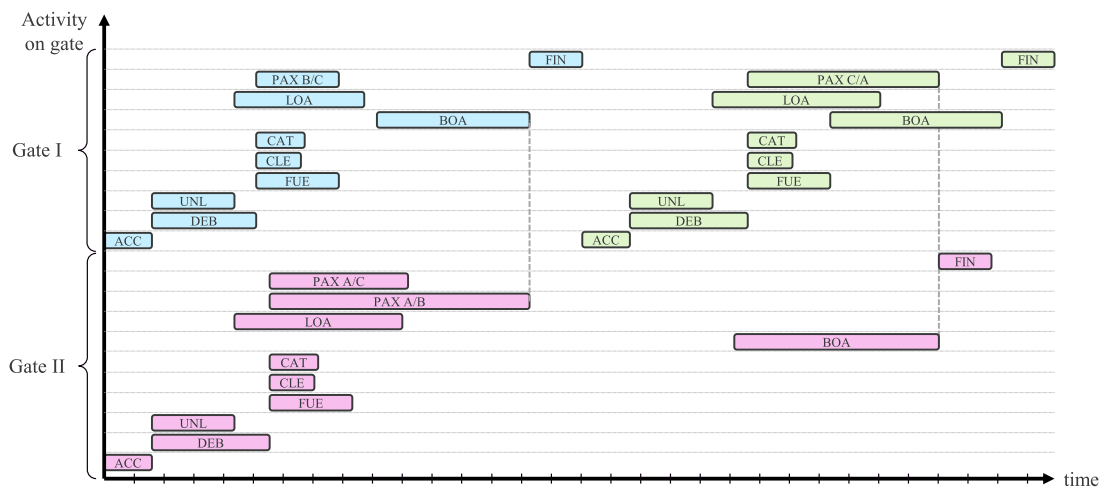


Fig. 5. An example of a turnaround schedule at an airport containing three aircraft and two gates with passenger connection between the aircraft.

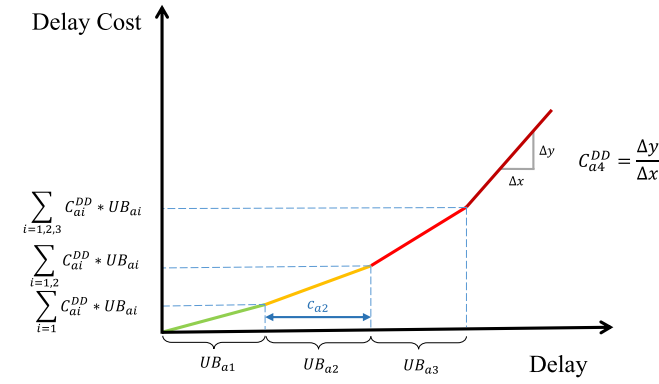


Fig. 6. Piece-wise cost function of departure delay for aircraft a adapted from Cook and Tanner (2015).

subsequently estimated turnaround time is a key task for airlines, especially in their hub airports.

As shown in Fig. 4a, we differentiate between the following sub-processes during the turnaround: Acceptance (ACC), the processes of accepting aircraft to be prepared for turning around, Unloading (UNL), unloading the baggage of arrival flight, Deboarding (DEB), passenger disembark, Catering (CAT), loading the catering equipment and supplement, Cleaning (CLE), cleaning the aircraft for departure flight, Fueling (FUE), refueling the aircraft depending on the next flight duration, Loading (LOA), loading the baggage of the departure flight, Passenger Transfer (PAX), transfer time between two aircraft in order to connect passenger connection, Boarding (BOA), passenger embarking, Finalization (FIN), the last process contains the last check of turnaround, closing doors and taxi out.

Since each (single) passenger transfer connects two aircraft, parallel turnaround operations, especially in hub airports, are challenging for airlines and airports. As depicted in Fig. 4b, each delayed flight in an airline network holds the potential to cause severe delays in the entire network due to passenger connections. Typical operational measures such as the cancellation of the connection, passenger compensation, and rebooking of the flight are the actions to overcome delays caused by passenger connections (see Fig. 4c).

3.2. Integration of gates and turnaround

Parallel (connected) turnaround operations get even more complicated when the aircraft gate assignment has to be considered. In this research, we consider both types of gates, namely remote stand and contact stand, with different travel times between each gate depending on the gates' position toward each other. Apparently, when all gates are occupied, aircraft have to wait until one gate is available. Furthermore, the mandatory minimum connection time depends on the location of the assigned gates. In other words, in contrast to other processes with a constant duration, the passenger transfer duration varies depending on the respective passenger connectivity.

Fig. 5 shows a simple case with three aircraft (A: pink; B: blue; C: green) and two available gates (gates I and II). There is a passenger connection from aircraft A to B, aircraft B to C, and Aircraft C to A in this concrete case. Therefore, the boarding process of the aircraft B can only be finished when the passenger of aircraft A reaches the new flight unless this connection is canceled. In the case of aircraft B and C, there is no setback because they have been sequentially ordered, and aircraft C must wait after its arrival until at least one gate is available. Analogously, the boarding process of aircraft A can also be finished after the passenger transfer process of aircraft C.

3.3. Mathematical formulation

To minimize network delays, it is necessary to optimally distribute resources and optimize the sequence of activities. The transfer time of passengers and the sequence of passenger arrivals depends mainly on the location of the assigned gates. Actual transfer times NTT_{kl} from stand k to stand l change for all transfer passengers to their respective onward flights when an incoming flight is assigned to a new stand, and accordingly for the passengers boarding that aircraft.

In addition, recovery actions are prepared to compensate for arising delays. One of the most common recovery options is the prioritization of aircraft servicing processes for aircraft A/C (τ) so that limited resources – for instance, cleaning crews – are assigned primarily to prioritized turnarounds to reduce the sub-process duration by the factor ω . Another convenient option is Arrival prioritization, which induces an additional workload for air traffic controllers to be more linked to resource availability than to cost. To reduce the delay of an arrival flight, the airline can request air traffic control to prioritize the approach or taxi-in process of aircraft a . Pilon, Cook, Ruiz, Bujor, and Castelli (2016) concluded that flight prioritization for optimal arrival sequencing might be transferred into the framework of the user driven prioritization process rather than being a stand-alone tactical control option for airlines.

In the operational situation, both the absolute and the average costs of delay increases over time (Cook & Tanner, 2015). The delay cost function can be divided into different increasing segments as shown in Fig. 6. In this paper, we implemented a convex piece-wise linear function with four segments used to describe the delay cost. The marginal cost of delays is a non-decreasing function of the delay. In this example of Fig. 6, the cost function is divided into four different delay segments with unique cost functions. For instance, if the departure delay, c_{a2} , falls within segment two (in UB_{a2} , upper bound of admissible delay of level 2) the delay cost is calculated by summation of the first segment constant cost ($C_{a1}^{DD} * UB_{a1}$) and second segment cost obtained by C_{a2}^{DD} times the remaining departure delay is located within segment two ($c_{a2} - UB_{a1}$).

In the following, the sets, parameters, and variables used in mathematical modeling are described, and the formulation of the model is accordingly developed. First, the primary model is developed based on a realistic operational situation in airports, and as the second step, recovery actions are integrated into the main model.

Sets:

A/C	set of all aircraft
TP	set of all turnaround sub-processes
AC	sub-set of TP, containing all aircraft acceptance processes
DE	sub-set of TP, containing all aircraft de-boarding processes
FU	sub-set of TP, containing all aircraft fuelling processes
CA	sub-set of TP, containing all aircraft catering processes
CL	sub-set of TP, containing all aircraft cleaning processes
BO	sub-set of TP, containing all aircraft boarding processes
UL	sub-set of TP, containing all aircraft unloading processes
LO	sub-set of TP, containing all aircraft loading processes
FI	sub-set of TP, containing all aircraft finalization processes
OB	sub-set of TP, containing all aircraft off-block processes
DS	set of piece-wise linear delay costs (Delay cost segment)
PA	set of all passenger transfer processes
ST	set of all aircraft stands
AP	set of dummy start and end nodes for sequencing of recovery resources (see the network in Fig. 4a)

Parameters:

C_{jn}^{DD}	cost of departure delay of aircraft j for delay level n
N^{AP}	number of available arrival prioritization
d_i	duration of turnaround process i
NTT_{kl}	needed transfer time between stands k and l
γ	time reduction factor achieved by flight prioritization
T	transition time between two stand assignments

(continued on next page)

(continued)

$SIBT_a$	scheduled in-block time for aircraft a
$EIBT_a$	estimated in-block time for aircraft a
$SOBT_a$	scheduled off-block time for aircraft a
UB_{as}	upper bound of admissible delay level s for aircraft a
RA_i	related aircraft of turnaround process i
M	big M parameter
PM	precedence matrix links to preceding and succeeding processes ($PM \subseteq TP \times TP$)
C_i^{CP}	cost of cancelled passenger transfer i

Variables:

s_i	starting time of process i
v_a	total departure delay for aircraft a
c_{as}	delay of aircraft a in delay segment s
$eobt_a$	estimated off-block time for aircraft a based on the scheduling process
τ_a	binary variable - equal to 1 if flight prioritization for aircraft a is assigned, and 0 otherwise
χ_{ak}	binary variable - equal to 1 if aircraft a is assigned to stand k , and 0 otherwise
ψ_{abkl}	binary variable - equal to 1 if aircraft a is assigned to stand k and aircraft b to stand l , and 0 otherwise
xS_{abk}	binary flow variable - equal to 1 if turnaround of aircraft b is done directly after turnaround of aircraft a on stand k , and 0 otherwise
κ_i	binary variable - equal to 1 if passenger transfer i is cancelled, and 0 otherwise

The proposed model is structured as follows:

$$\min \sum_{a \in A} \sum_{s \in DS} C_{as}^{DD} c_{as} \quad (1)$$

$$s_i \geq EIBT_a - \gamma \tau_a \quad \forall i \in AC, RA_i = a \quad (2)$$

$$s_i \leq SOBT_a + v_a \quad \forall i \in OB, RA_i = a \quad (3)$$

$$v_a = \sum_{s \in DS} c_{as} \quad \forall a \in A / C \quad (4)$$

$$c_{as} \leq UB_{as} \quad \forall a \in A/C; \forall s \in DS \quad (5)$$

$$s_j \geq s_i + d_i \quad \forall i \in \{AC, FI\}; \forall j \in TP | (i, j) \in PM \quad (6)$$

$$s_j \geq s_i + NTT_{kl} \chi_{ak} \chi_{bl} \quad \forall i \in PA; \forall j \in BO | (i, j) \in PM, RA_i = a, RA_j = b; \forall k, l \in \{ST\} \quad (7)$$

$$\sum_{a \in \{A/C, AP\}} xS_{abk} = \chi_{bk} \quad \forall b \in A / C; \forall k \in \{ST\} \quad (8)$$

$$\sum_{b \in \{A/C, AP\}} xS_{abk} = \chi_{ak} \quad \forall a \in A / C; \forall k \in \{ST\} \quad (9)$$

$$s_i \geq eobt_a + T - M(1 - xS_{abk}) \quad \forall a \in A/C; \forall b \in \{A, AP\}; \forall i \in AC, RA_i = b; \forall k \in \{ST\} \quad (10)$$

$$\sum_{k \in \{ST\}} \chi_{ak} = 1 \quad \forall a \in A / C \quad (11)$$

$$\sum_{a \in A/C} \tau_a \leq N^{AP} \quad (12)$$

The objective function (1) aims to minimize the total cost caused by delay occurred. Delay occurs when the arrival delays arise and when operations cannot begin because of resource absences. Constraint (2) determines the earliest start time of each turnaround process, which can commence after its respective estimated in-block time. If the calculated Off-Block time exceeds the scheduled Off-Block time, the difference, called delay, is calculated by (3). Since the calculation of delay cost is based upon a piece-wise function, Constraints (4) and (5) calculate the total delay as a summation of the delay segments and assure that each segment is limited to an upper value determined as a maximum delay time in that segment. Scheduling constraints (6) ensure that each sub-process can only start once all preceding processes have been finished. Constraints (7) states that the boarding process can only start when the passenger connection process, which depends on the stand allocation of inbound and outbound flights and is calculated by needed transfer time NTT_{kl} , has been taken place. Routing resource constraints (8)–(10) consider that a turnaround operation can only begin when the respective resource is available and ensure that all stands can only be occupied by one aircraft at a time. T induces the transition time between two subsequent assignments. Constraint (11) states that each aircraft must be allocated to one of the existing gates. Constraint (12) limits the total number of available arrival prioritization.

Since the gate assignment models belong to quadratic mathematical problems in general, Constraint (7) would be re-written by Constraints (13)–(16). This ensures that the problem remains linear and is capable of being solved by linear solution approaches.

$$s_j \geq s_i + NTT_{kl} \psi_{abkl} \quad \forall i \in PA; \forall j \in BO | (i, j) \in PM, RA_i = a, RA_j = b; \forall k, l \in \{ST\} \quad (13)$$

$$\chi_{ak} + \chi_{bl} \leq 1 + \psi_{abkl} \quad \forall a, b \in A/C; \forall k, l \in \{ST\} \quad (14)$$

$$\chi_{ak} \geq \psi_{abkl} \quad \forall a, b \in A/C; \forall k, l \in \{ST\} \quad (15)$$

$$\chi_{bl} \geq \psi_{abkl} \quad \forall a, b \in A/C; \forall k, l \in \{ST\} \quad (16)$$

3.3.1. Cancellation of passenger connections

Outbound steering goes hand in hand with inbound steering, as aircraft which have no initial arrival delay can be held in position to avoid miss-connections. At the same time, the airline might also decide to cancel the connection when the delay in the network is severe (see Fig. 4) (Pternea & Haghani, 2018). One of the most common options to bring flights back to schedule is canceling passenger connections in exchange for related costs depending on various factors such as passenger care, re-booking, lodging, and compensation. The cost C_i^{CP} for cancelling passenger transfer i are adopted from Cook and Tanner (2015).

The total passenger miss-connection cost is calculated by (17) consists of the cost of breaking the connection i which takes effect when κ_i is one. In this case, the boarding process of the successor flight (outbound flight) can begin without waiting for connecting passengers. Constraint (7) should be re-written by Constraint (18) that exercises κ_i to break passenger connection i .

$$\sum_{i \in PA} C_i^{CP} \kappa_i \quad (17)$$

$$s_j \geq s_i + NTT_{kl} \psi_{abkl} - M \kappa_i \quad \forall i \in PA; \forall j \in BO | (i, j) \in PM, RA_i = a, RA_j = b; \forall k, l \in \{ST\} \quad (18)$$

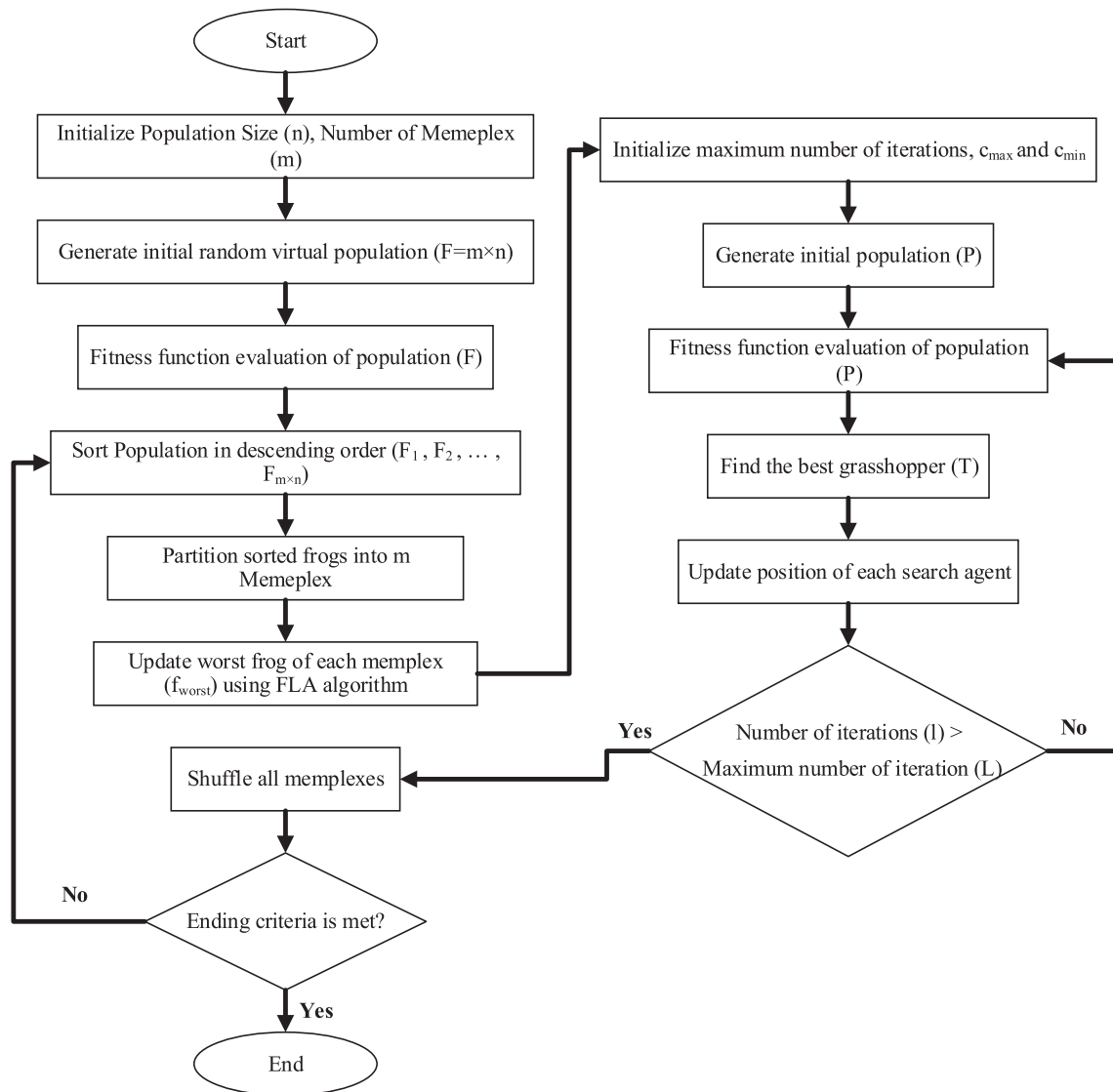


Fig. 7. Flowchart of Hybrid SFLA-GOA algorithm.

4. Solution method

As described in Section 1, the combination of RCPSP and Gate Assignment problems belongs to NP-Hard problem class. Ergo, there is a need to develop a solution approach to tackle this kind of hard computational problems. Chu, Xu, and Li (2019) developed a resource-constrained project scheduling problem with overlapping activities. Heuristic algorithms were developed based on the roll-out policy. These algorithms are capable to integrate with Genetic Algorithms (GA), Simulated Annealing (SA), and Tabu search (TS). To find a solution for Gate Assignment models, which can also be considered as Job-Shop scheduling problems that consider Gates as Machines and Turnaround activity as Jobs, meta-heuristic algorithms can be implemented. Wang, Li, Chen, and Mao (2020) improved two meta-heuristic algorithms, GA and SA, to solve parallel machine scheduling problems on a large scale for two MIP problems. Lei, Li, and Wang (2019) developed, based on variable neighborhood search and imperialist competitive algorithm, a two-phase meta-heuristic algorithm to solve a flexible job-shop scheduling problem while considering consumption threshold in the constraints.

4.1. Shuffled Frog-Leaping Algorithm (SFLA)

Shuffled Frog-Leaping Algorithm (SFLA) is introduced by Eusuff, Lansey, and Pasha (2006). This algorithm uses mimics to communicate between frogs to exchange their experience of the best location with maximum food. In this algorithm, frogs compose to build a particle swarm to search into the solution space. To make this happen, virtual frogs (i) are divided into some memplexes based on their ranks in the entire population using (19), with a higher probability for the best frogs to be chosen for comprising the sub-memplexes. Each frog perceives the experience of other frogs and evolves it into a process called memetic evaluation. The local improvement is made simultaneously in each memplex, and then all of the experience is shuffled to form a whole population to ensure global exploration. If a frog cannot find better results during the evaluation process, it is substituted through a mechanism called "Censorship" to explore into random search spaces (exploration). The process of updating each frog meme is done by (20) where P_W is the current position of the frog, and P_B is the position of the best frog within the same memplex. If Eq. (20) fails in producing better result, Eq. (21) is used where P_X presents the position of the best ever found frog.

$$p_i = 2(n+1-i)/n(n+1) \quad i = 1, \dots, n \quad (19)$$

$$P_W^{new} = P_W + \text{int}[\text{rand}(P_B - P_W)] \quad (20)$$

$$P_W^{new} = P_W + \text{int}[\text{rand}(P_X - P_W)] \quad (21)$$

4.2. Grasshopper Optimization Algorithm (GOA)

Saremi, Mirjalili, and Lewis (2017) developed Grasshopper Optimization Algorithm (GOA), which simulates grasshopper behavior in nature. This memetic algorithm models memetic grasshopper swarm to find the solution to optimization problems. Exploration and exploitation in this algorithm are modeled by repulsion forces and attraction forces between grasshoppers, respectively. The position of grasshopper i (X_i) depends naturally on social interaction with other grasshoppers (S_i), the gravity force of grasshopper i (G_i), and the wind direction (A_i), as depicted in (22). To make this behaviour capable to solve optimization

problems, Eq. (22) is modified as (23), where ub_d and lb_d are upper and lower bounds, respectively, and $s(r)$ is the social interaction function, $\widehat{T}_d$ is the best so far solution, and c is the coefficient of decreasing comfort zone moving toward exploration rather than exploitation, calculated by (24).

$$X_i = S_i + G_i + A_i \quad (22)$$

$$X_r^d = c \left(\sum_{j=1, j \neq r}^{N_s} c \frac{ub_d - lb_d}{2} s \left(\left| x_j^d - x_r^d \right| \right) \frac{x_j^d - x_r^d}{d_{rj}} \right) + \widehat{T}_d \quad (23)$$

$$c = c_{max} - l \left(\frac{c_{max} - c_{min}}{Iter_2} \right) \quad (24)$$

4.3. Hybrid of SFLA and GOA (SFLA-GOA)

In a two-level decision-making environment, here gate assignment and sequencing of activities, the search space should deeply explore to assure reaching a better solution during the search in the solution space (Kacem, 2013). There are strong possibilities that SFLA algorithm gets stuck at a local optimum point. Furthermore, the convergence of the algorithm is slow (Niknam, Narimani, & Azizipanaah-Abarghoee, 2012; Sharma, Sharma, & Sharma, 2017). On the other hand, when the P_W and P_X are near each other, the modification of the frog (P_W^{new}) is not great that retards the improvement of the solution. Additionally, the meme information of the best frog (P_B) is used only once in each improvement (Kaveh, Talatahari, & Khodadadi, 2019). GOA updates each particle's position based on its current position, positions of all other particles, and the global best particle. Ergo, to overcome the shortcomings of SFLA, Hybrid SFLA-GOA is developed in this paper. Hybridization of meta-heuristics algorithms (i.e., GA combines with PSO) is an interesting field that can give us more insights regarding the behavior and potential advantages and disadvantages of different algorithms (Sangsawang, Sethanan, Fujimoto, & Gen, 2015; Zaman, Elsayed, Sarker, & Essam, 2020).

The proposed algorithm benefits the advantage of SFLA algorithm to swiftly find solution regions to find local optimum points. Afterward, the GOA is used to exploit these regions to find global solutions. This algorithm is executed as depicted in Fig. 7.

First, the initial population is generated, and the function value of each search agent is calculated separately by the objective cost function. Next, a part of the population is selected to be partitioned into memplexes. In the next stage, Frog-Leaping Algorithm (FLA) is implemented to search locally within each memplex. If the memplexes result is not improved, the search agent will be updated according to the global best agent (so far). In case of failure, the search agent will be replaced by a new random agent. Next, all of the memplexes are considered as a new population for GOA algorithm to seek a global solution, shuffle the population, and compare it with SFLA algorithm. If the positions of agents, obtained by GOA, are improved than the initial population, they replace and form the new population in the next iteration of SFLA. If the new population is successfully replaced, the global best solution found so far will be swapped accordingly. Lastly, if termination criteria are not met, the algorithm starts once again. Otherwise, it stops, and the last global result should be considered as the final result. Algorithm 1 shows the pseudo-code of the proposed hybrid SFLA-GOA algorithm.

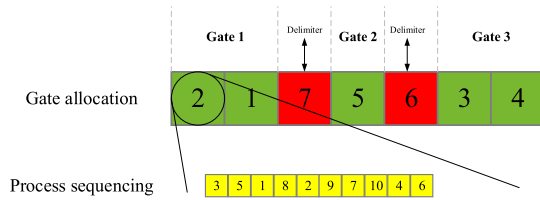
Algorithm 1. Hybrid SFLA-GOA algorithm Pseudo Code

Algorithm 1: Hybrid SFLA-GOA algorithm Pseudo Code

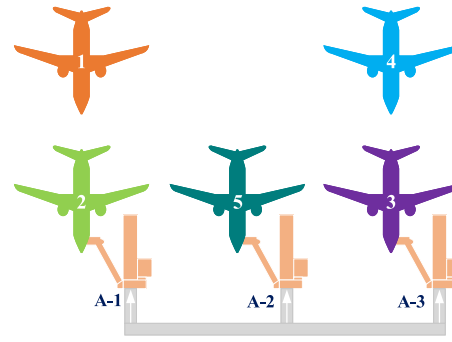
```

1 Set initial parameters for SFLA ( $m, n, q, N_1, Iter_1$ )
2 Generate randomly a virtual population of frogs ( $m \times n$ )
3 while  $i \leq Iter_1$  do
4   Sort population in descending order
5   Partition all frogs (whole population) into  $m$  memeplexes
6   Determine best frogs ( $P_b$ ) and worst frogs ( $P_w$ ) within each memeplex
7   for  $j \leftarrow 1$  to  $m$  do
8     for  $k \leftarrow 1$  to  $N_1$  do
9       Construct a sub-memeplex of frog ( $q$ ) with chance calculated by (20)
10      Improve the position of the worst frog ( $P_W^{new}$ ) by the best frog ( $P_B$ ) within
          memeplex  $m$  using (21))
11      if  $P_W^{new} \leq P_W$  then
12         $P_W \leftarrow P_W^{new}$ 
13      else
14        Calculate  $P_W^{new}$  by the best ever found frog ( $P_X$ ) using (22)
15        if  $P_W^{new} \leq P_W$  then
16           $P_W \leftarrow P_W^{new}$ 
17        else
18          Replace the worst frog's position with a new random position ( $P_W^{new}$ )
19        Upgrade the memeplex ( $m$ )
20 Set initial parameters for GOA ( $Iter_2, c_{min}, c_{max}, N_2$ )
21 Initialize the swarm of grasshoppers ( $X$ )
22 Determine best frog in whole population ( $\hat{T}$ )
23 while  $l \leq Iter_2$  do
24   Update parameter  $c$  by (25)
25   for  $r \leftarrow 1$  to  $N_2$  do
26     Normalize the distance between grasshoppers to [1,4]
27     Calculate social interaction of grasshopper  $r$  with other grasshoppers in swarm ( $X$ )
28     Update the position of grasshopper  $r$  ( $X_r^d$ ) using (24)
29     if  $X_r^d \notin [var_{min}, var_{max}]$  then
30       Bring the grasshopper position back
31     Update best ever found solution ( $\hat{T}$ )
32    $i \leftarrow i + 1$ 
33 Return best ever found solution ( $\hat{T}$ )

```



(a) Solution string depicts assignment of aircraft to the gates



(b) Schematic representation of the example solution

Fig. 8. Solution presentation for the case of 5 aircraft on 3 gates.**Table 1**

Different levels of initial parameters to run SFLA algorithm.

Row	Parameters	Levels		
		Low	Medium	High
A	Iter ₁	20*	40	60
B	n	35	50	70*
C	m	3*	5	8
D	q	30%	50%*	60%
E	N ₁	5	10*	20

* Chosen level.

Table 2

Different levels of initial parameters to run SFLA-GOA algorithm.

Row	Parameters	Levels		
		Low	Medium	High
A	Iter ₁	20*	40	60
B	n	35	50	70*
C	m	3*	5	8
D	q	30%	50%*	60%
E	N ₁	5	10*	20
F	Iter ₂	3*	5	10
G	N ₂	20*	30	40
H	C _{max}	1*	10	50
I	C _{min}	–	10 ⁻⁵ *	–

* Chosen level.

4.4. Solution presentation

In this section, a solution representation is introduced to be used for both SFLA-GOA and SFLA algorithms. In the first phase, aircraft $\{1, 2, \dots, I\}$ should be assigned to the gates $\{1, 2, \dots, J\}$. For this purpose, a string

is considered in which the numbers greater than I represent as delimiters (red boxes) and the numbers between (green boxes) are the sequence of the aircraft assigned to that gate (see Fig. 8). In this case, two aircraft with a passenger connection are assigned to a gate, and this connection should be canceled at a specific cost.

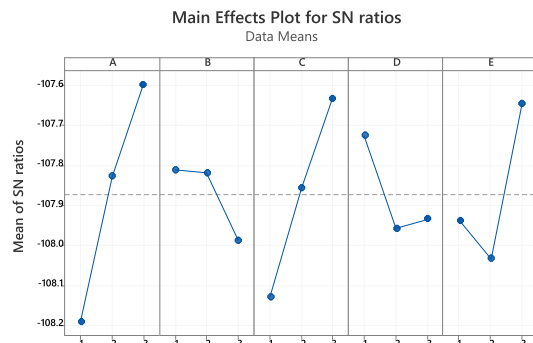
4.5. Solution methods evaluation

To evaluate the performance of the proposed Hybrid method and SFLA method in solving the problem, a set of standard test problems is considered in Table 3. In this table, the size of the problems increases with the number of aircraft and gates. To make the result of the proposed SFLA-GOA and SFLA algorithms comparable, Taguchi experimental design (Taguchi, Chowdhury, & Wu, 2005) is used to parameter tuning of both algorithms. To make this happen, three different levels for each parameter are defined, and the most appropriate level for each parameter is selected by the analysis of experiments in Minitab® Statistical Software. Tables 1 and 2 show the level of parameters and the selected one for each parameter and Figs. 9 and 10 exhibit the result of the tuning process.

The model was coded in the software Matlab 2018b and was run on a system with Windows 10-64bit OS, 24-kernel, and 32 GB of RAM. The results in Table 3 were obtained with maximum solution time related to each test problem, which has been considered as a termination criteria to make a fair comparison between obtained solutions. To show the performance of algorithms, two criteria, namely %PRAS (Percentage Reduction in the Average Solution) (25) and %PRBS (Percentage Reduction in the Best Solution) (26) which compare the distance between the average and best solution of both algorithms, respectively. Regarding the performance algorithm, Fig. 11 shows that SFLA-GOA algorithm outperforms SFLA algorithm in terms of the quality of the



(a) Mean of Means plot



(b) Mean of Signal-to-noise (SN) ratio (smaller is better)

Fig. 9. SFLA parameter tuning.

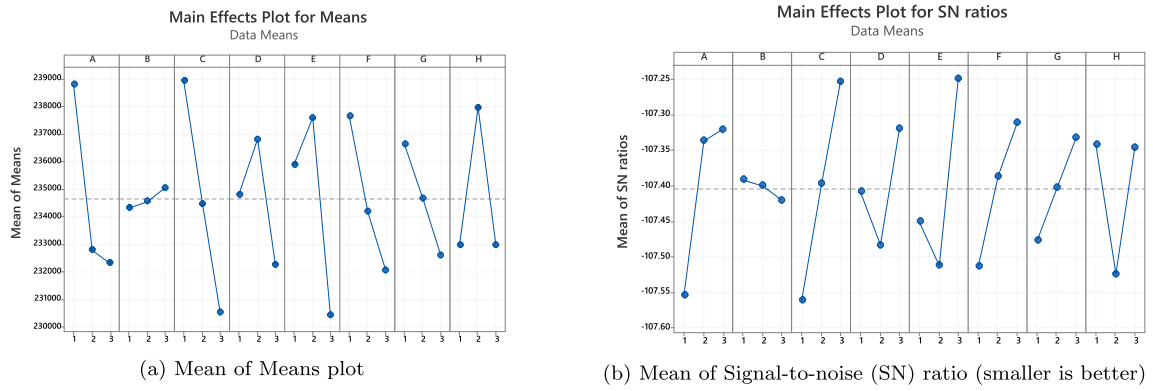


Fig. 10. SFLA-GOA parameter tuning.

solution. For instance, there is no distinct difference in small test problems, but the level of performance increases in bigger problems, 14.20% in average solution and 14.97% in terms of best-found solution (Fig. 11a). As can be seen in Fig. 11, the average cost found by Hybrid SFLA-GOA is continuously below the average cost obtained from SFLA algorithm.

$$PRAS = \frac{(Best_{SFLA-GOA} - Best_{SFLA})}{(Best_{SFLA-GOA})} \times 100 \quad (25)$$

$$PRBS = \frac{(Best_{SFLA-GOA}^{Ave} - Best_{SFLA}^{Ave})}{(Best_{SFLA-GOA}^{Ave})} \times 100 \quad (26)$$

5. Case study

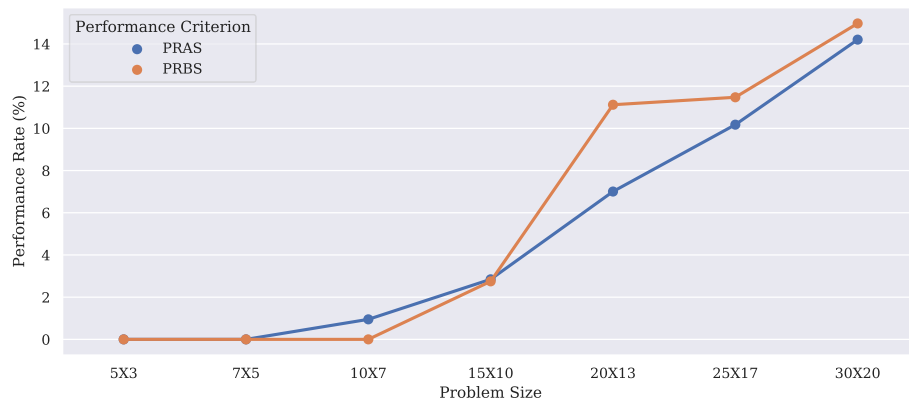
We use the Frankfurt airport environment for our case study, including gate positions, terminal layout, and flight plan information. This airport was handling 70 Mio. passengers and 514k aircraft

Table 3

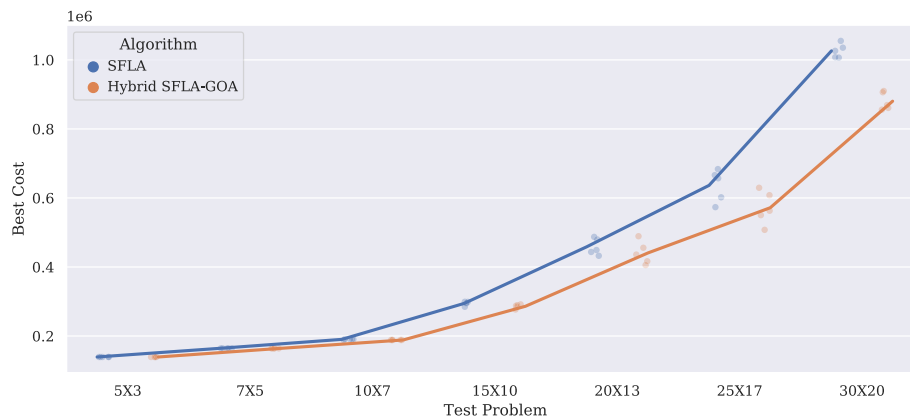
Results of solving the test problems using different problem size, comparing SFLA and SFLA-GOA algorithms in terms of average and best costs.

Problem Size*	Time	Iteration	SFLA			SFLA-GOA		
			Solution	Average	Best	Solution	Average	Best
5 × 3	125.5s	1	139235	139235	139235	139235	139235	139235
		2	139235			139235		
		3	139235			139235		
		4	139235			139235		
		5	139235			139235		
7 × 5	194.7s	1	164305	164305	164305	164305	164305	164305
		2	164305			164305		
		3	164305			164305		
		4	164305			164305		
		5	164305			164305		
10 × 7	230.3s	1	190580	190301	188495	188495	188495	188495
		2	190925			188495		
		3	188495			188495		
		4	190580			188495		
		5	190925			188495		
15 × 10	235.4s	1	295160	290990	276905	291650	289840	285545
		2	299000			276905		
		3	299150			285545		
		4	294995			287625		
		5	284735			289385		
20 × 13	900.6s	1	443506	440607	405734	416583	458451	432448
		2	479840			489136		
		3	432448			455645		
		4	449222			435937		
		5	487240			405734		
25 × 17	1027.0s	1	666174	636434	573412	507624	571677	507624
		2	683637			562978		
		3	601897			550209		
		4	573412			608066		
		5	657051			629508		
30 × 20	1903.5s	1	1007730	1026211	1006664	906315	880442.4	855940
		2	1026211			909930		
		3	1035392			869141		
		4	1006663			855940		
		5	1055056			860886		

* Number of aircraft × Number of available gates



(a) PRAS and PRBS performance criteria



(b) Average of best costs

Fig. 11. Comparison of the performance of algorithms for each problem size in terms of average cost and performance criteria.

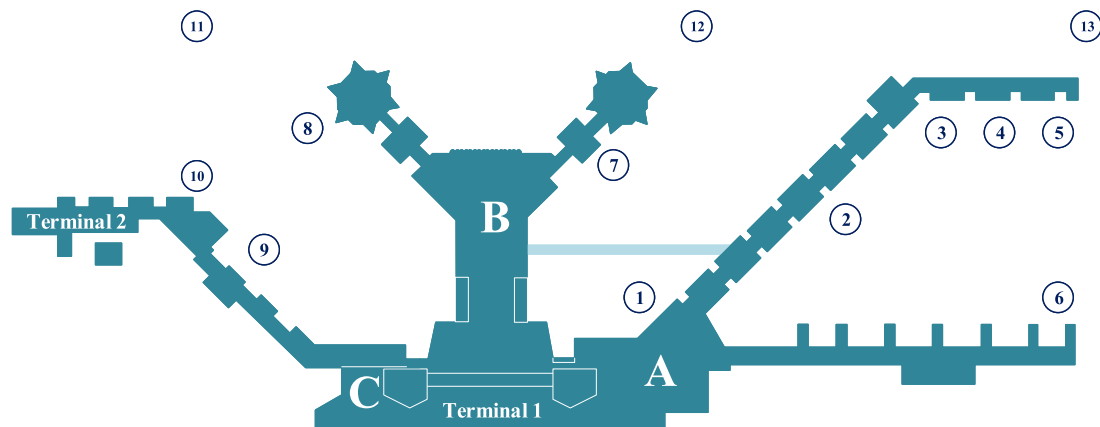


Fig. 12. Simplified environment of Frankfurt Airport with 13 aircraft stand positions used in the case study.

movements in 2019, with the runway capacity coordinated to 104 aircraft movements per hour (take-off and landings). Lufthansa Airlines, a network carrier, operates one of its central hubs at Frankfurt Airport and handles many passenger transfers. Fig. 12 exhibits the general layout of the airport environment and the location of 13 gates used in the case study. In our example, the gates were selected so that the respective position of the gates also leads to a significant influence in the optimization. Transfer passengers will not only arrive from national

feeder flights but also from continental and intercontinental flights.

The operational scenario is defined by 20 aircraft flying from and to different airports (airports abbreviated by IATA codes). In Table 4 the flights are emphasized by both aircraft and passenger connection. The passenger transfer matrix has to be read as follows: the first aircraft arrives from EWR (Newark Liberty International Airport) at 7:30 and will depart at 10:45 to ORD (O'Hare International Airport), where 45 transfer passengers will continue their travel to AMS (Amsterdam

Table 4

Passenger transfer matrix between flights using 20 aircraft with Scheduled In-Block Time (SIBT), Scheduled Off-Block Time (SOBT), and airport and passenger connections.

-	SIBT	ORD	AMS	DME	TXL	JFK	MLA	MUC	OTP	AMS	LED	MAD	CDG	LHR	SCQ	TLV	CPH	STO	LIN	EDI	VIE
SOBT		10:45	8:20	8:35	8:45	11:00	9:05	9:15	9:30	9:15	9:35	9:30	9:30	10:00	10:00	10:20	9:55	10:05	10:55	11:00	10:50
EWR	7:30	-	48	25	21	-	4	4	4	-	4	-	-	-	-	5	-	12	2	-	15
HAM	7:40	3	-	20	-	4	2	6	5	-	8	4	3	-	1	3	-	-	3	2	-
LED	7:50	-	-	-	-	39	30	-	-	-	-	8	14	-	7	-	-	-	5	-	-
TXL	7:55	11	-	-	-	16	13	-	-	16	-	10	5	9	3	-	-	-	2	6	-
MIA	8:00	-	-	-	44	-	21	7	35	-	33	-	-	-	-	7	38	35	10	-	8
MUC	8:00	5	-	-	5	1	-	-	-	23	-	5	16	3	4	-	14	4	-	4	-
HEL	8:10	15	-	-	-	9	6	14	-	-	-	7	16	10	9	-	-	-	6	-	-
OSL	8:10	-	-	-	-	-	-	15	20	-	-	9	17	-	1	15	-	-	-	-	6
STO	8:15	-	-	-	-	8	3	24	-	-	-	8	3	16	13	1	-	-	10	-	-
ATH	8:25	15	-	-	-	3	-	-	-	4	-	13	12	10	13	-	5	1	-	3	-
DME	8:25	2	-	-	-	27	-	-	-	24	-	15	7	4	8	-	-	-	6	12	-
MAD	8:40	-	-	-	-	-	-	-	10	-	24	-	-	-	-	10	18	3	-	-	15
MAN	9:00	-	-	-	-	-	-	-	-	-	-	-	-	-	9	44	-	-	9	-	28
LYN	9:00	24	-	-	-	-	-	-	-	-	-	-	-	-	-	-	8	27	-	11	12
VLC	9:00	-	-	-	-	-	-	-	-	-	-	-	-	-	-	19	16	10	-	13	10
MLA	9:10	34	-	-	-	33	-	-	-	-	-	-	-	5	-	-	2	2	-	10	-
ZRH	9:20	12	-	-	-	9	-	-	-	-	-	-	-	-	-	-	-	18	-	28	-
BRU	10:05	-	-	-	-	7	-	-	-	-	-	-	-	-	-	-	-	-	18	-	17
HAM	10:10	-	-	-	-	17	-	-	-	-	-	-	-	-	-	-	-	-	28	3	6
VIE	9:40	33	-	-	-	29	-	-	-	-	-	-	-	-	-	-	-	-	-	6	-

Table 5

Result obtained for each scenario day - actual In-Block and Off-Block times.

Scenario	Solution Gap*	Upper Bound	SFLA-GOA	SFLA-GOA	Improvement
(Day)	(Gurobi Solver)	found	Solution	Run Time (Sec.)	in solution
Day 1	27.15%	56,637	48,238	1027.14	14.83%
Day 2	35.20%	41,174	32,165	1052.02	21.88%
Day 3	49.51%	333,805	204,422	1051.45	38.76%
Day 4	25.60%	56,150	48,620	1014.10	13.41%
Day 5	24.21%	84,817	68,447	966.53	19.3%
Day 6	26.66%	131,395	112,776	1014.12	14.17%
Day 7	27.24%	66,484	55,348	1023.56	16.75%
Day 8	30.02%	86,956	73,278	1040.39	15.73%
Day 9	30.39%	122,072	107,191	995.99	12.19%
Day 10	30.73%	226,251	195,739	1039.74	13.49%

* Results are obtained within 3,600 s run time (see Fig. 2).

Airport Schiphol), 25 to DME (Moscow Domodedovo Airport), 21 to TXL (Berlin Tegel Airport), and so forth. The second aircraft arrives from HAM (Hamburg Airport) and departs to AMS, the third aircraft arrives from LED (Pulkovo Airport) and departs to DME, and so forth. The number of transfer passengers was exemplarily adjusted (derived from airline information) and adapted according to the required minimum connecting times for passenger transfer at Frankfurt Airport. At this point in time, we do not include tactical decisions of the airlines, such as rerouting aircraft to other destinations. This will be part of future research activities.

In Section 4, we showed a better performance of the proposed hybrid algorithm in solving the mathematical model. Thus, we used the Hybrid SFLA-GOA Algorithm to solve ten different scenarios (days of operations) extracted from Lufthansa's 2017 summer flight schedule at Frankfurt Airport. These days differ by the actual times measured for aircraft arrival and departure, which are also different from the initial

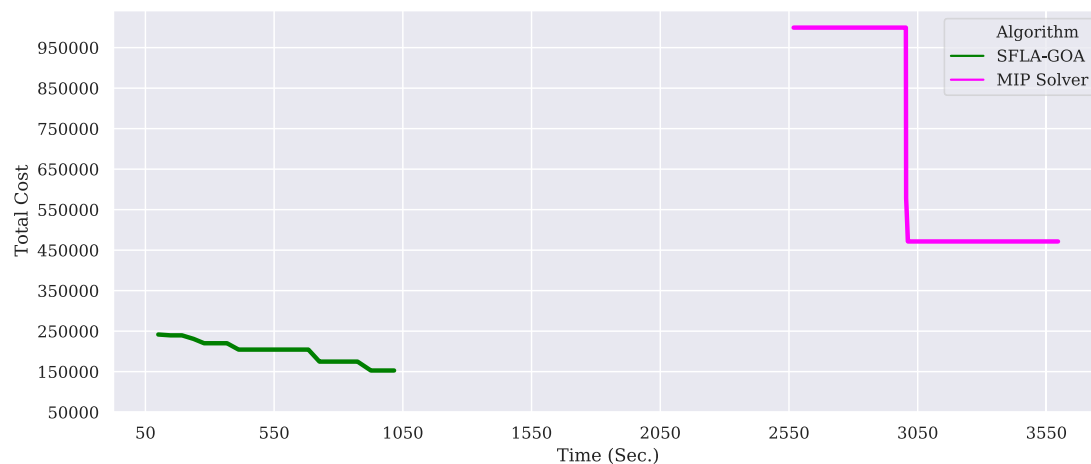


Fig. 13. The comparison of solution time between Hybrid SFLA-GOA Algorithm and MIP solver (scenario "Day 3").

flight schedule. All scenarios have a mean arrival delay above the average arrival delay considering all operational days in the corresponding period (7:30 to 11:00).

The mathematical model was solved using Gurobi package 8.0 on a 24-kernel CPU with 32 GB RAM on Windows 10. The solution process was aborted after 90 min in case the optimality of a solution could not be proven, and there was still a gap between the upper and lower solution bounds (see Table 5). Our case study implemented a standard case consisting of 20 aircraft and 13 gates, which can lead to rapidly increasing complexity with an additional number of parallel, coupled turnarounds. Table 5 provides a comparison between proposed algorithm with the MIP solvers. The quality of results in terms of time and achieved objective is improved from almost 12% up to 40% in some cases with severe arrival delay in the system. Solution gaps are the gaps between the lower bound and upper bound found after one hour run time. For instance, 27.15% solution gap in scenario "Day 1" means the gap between the upper bound and lower bound is 27.15%, with upper bound 56,637 and lower bound 41,260. Since the solver could not find an optimal solution in an appropriate time, the upper bound can be used as the best solution so far. The solutions, found using Hybrid SFLA-GOA algorithm and the corresponding time, are shown in the table. The last column in the table illustrates how many percentages of improvement in each scenario have been achieved by the proposed algorithm. Regarding the solution time, Fig. 13 illustrates the computation time for scenario "Day 3", which possesses the most significant solution gap. The first solution found by MIP solver was achieved after 2,550 s, while the calculation of SFLA-GOA algorithm was finished after 1,050 s.

6. Conclusion

Operational planning at the airport is challenged by deviations in aircraft operations (delay), local airport capacities (arrival and departure flows), local weather conditions, or resource availability. In our contribution, we integrated aircraft turnaround time optimization and gate assignment into one optimization model. We present a hybrid metaheuristic algorithm based on Shuffled Frog-Leaping Algorithm (SFLA) and Grasshopper Optimization Algorithm (GOA) to solve these NP-Hard problems. We derived input data for a case study from airport ground operations at Frankfurt Airport using the network carrier Lufthansa as an example airline. We are building up an operational scenario with the 20 flight and 13 available gates where transfer passengers connect the flights. For each scenario, flight plan information provides actual arrival and departure times, which differs from the initial flight schedule due to operational delays.

To overcome the drawback of SFLA algorithm, we develop a new hybrid SFLA-GOA approach that outperforms the SFLA algorithm in terms of the quality of the solution and solution time in different problem sizes. Two criteria, namely %PRAS (Percentage Reduction in the Average Solution) and %PRBS (Percentage Reduction in the Best Solution), are used to analyze the betterment of the solution and the run time. The quality of results in terms of the achieved objective is improved from almost 12% up to 40% in some cases with severe arrival delay in the system. The solution time using the current hardware environment is about twenty minutes.

Different stakeholders jointly work on integrated decision-making in the airport environment to achieve system-wide performance targets, such as maximum aircraft flow, minimum delay, or reaching target values for costs. In our future research, we will focus on tactical airline behavior by developing objective functions that represent the conflicting perspectives in the airline decision-making system. Hence, the algorithm can be extended to build and analyze Pareto frontier to deliver reliable solutions (Kizilay, Tasgetiren, Pan, & Süer, 2019). Furthermore,

the model scope can be broadened considering large-scale airline networks, such as airline alliances. Other swarm-based meta-heuristic algorithms, such as Grey Wolf Optimization and Krill Herd Algorithm, can be adapted to the proposed model to analyze the impact of a different combination of algorithms on the proposed model.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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